

petro, oil and gas, food and drinking water. And the scenario is changing fast as, like the developed nations, India is also aiming at enhancing the quality of its products and provide the market-conscious consumers a variety of choices. For that, it is looking up to industrial automation. Automation has many advantages besides being a solution to improve quality along with the environmental balance, such as the growing interface, quality control, optimisation and product tracking.

"Solutions capability is the key driver. Companies established in India are doing a lot of value addition in terms of application engineering and application software, including embedded software, other automation support services like consulting, project engineering, simulation, commissioning, AMC, education and training. This has created comfort and confidence in the eco-system about sustaining the drive for modernisation and adapting to the latest technologies," says Wadhwa.

In India, the success lies in offering solutions that can tackle real problems, and those manufacturers who follow this are doing well. "Industrial automation market worldwide is being driven by several trends. In emerging nations like India, industrial and factory automation is rapidly advancing. More and more manufacturing is getting automated, and these emerging countries are skipping generations to go with the latest available technologies. Integration to enterprise systems, sustainability and eco-friendliness and availability of smarter technologies like sensors, HMI and communication technologies are driving automation markets worldwide," says Somsubhro Pal Choudhury, managing director, Analog Devices India (ADI).

"Local Indian players struggle in terms of economies of scale, which is a distinct advantage enjoyed by the global players who have huge R&D expenditure along with a strong sales and marketing presence in the Indian market. Frugal innovation to reduce the total cost of ownership would

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definitely drive growth. Local players could also look for potential partnerships with the growing number of Indian design houses that are doing a lot of product development and designs for their global clients," Choudhury says.

Another factor driving the growth in this sector is government's thrust towards boosting the electronics manufacturing industry, especially the LED lighting and solar sectors. The increased manufacturing in these sectors will lead to increase in the number of automated plants, whereas the rising awareness about the rising energy costs is also driving the growth of the sector.

However, Tony Lennon, MathWorks industry manager for industrial automation, feels the government needs to do a little more to boost the industrial automation sector in the country. "The government needs to recognise that industrial automation technologies are a big enabler of productivity improvement, safety assurance, energy efficiency and environmental protection. If our industries were mandated by legislation to conform to better self-regulation, 100 per cent transparency and real-time audit, our technologies would be able to demonstrate the big difference that arises between 'automated for conformance' systems and otherwise," Tony adds.

Challenges faced. Talking about the challenges faced by the industry, Wadhwa says, "Increased productivity is one outcome of a series of benefits that automation brings to the table, or in this case the shop floor. Most adopters

are recognising that there are internal barriers to cross, and that does not come easy. Hence, automation must always be seen in the context of the bigger picture."

Wadhwa feels that automation is not just the replacement of man by machine to do routine tasks. It is the standardisation of processes and methods that can be fed into machines unlike humans. Of course not to forget machines that can be put to work in high-risk/hazardous areas, which we will not risk with human resources. "However the biggest benefit I consider is the possibility of training the very staff who were doing repetitive tasks in better, meaningful jobs which gives them, too, a chance to rise up the social ladder. In other words, it is a holistic processing tool towards sustainability. When our industry awakens to such a vision, people will embrace automation for all such benefits, not merely by an ROI approach," he adds.

But Tony thinks that it will become increasingly difficult to compete as the low-cost player in the market. "The need for greater efficiency and safety in machines and systems will outweigh the cost factor. Domestic Indian companies will need to offer a similar level of sophistication as their multinational competition. As the software content on their systems grows, they will need to adapt their development processes to meet the challenge," he added.

Innovations and new trends

"Machine builders are seeing that the complexity of the software running on their equipment continues to grow. It is a similar trend first seen by the automotive industry in the 90's. Improving efficiency and safety is being achieved through embedded software running on advanced industrial controllers. You can find more and more complex software written in IEC 61131 structured text and ANSI/ISO C code running on machines to provide the performance demanded by the market. MathWorks is supporting the challenge of developing more complex software by providing simulation tools